Prove the fact, not your ID.

Zero-knowledge credentials from the IC chip in Japan's My Number Card, verified in any browser and on Solana.

Yu Ayato



THE PROBLEM · WORLDWIDE

ID documents are leaking everywhere.

US & CANADA · SEP 2026

150M+

records from an ID-verification vendor: driver's licences, passport numbers

JAPAN · SEP 2026

1.6M

ID images from a car-sharing service's sign-ups

GLOBAL · OCT 2025

~70k

government ID photos sent for age-check appeals on a chat platform

US · JUL 2025

13k

selfies and photo IDs uploaded to verify accounts on a dating-safety app

AUSTRALIA · MAR 2023

7.9M

driver's licence numbers (and 53k passport numbers) from a consumer lender

Every one was a copy someone was asked to upload.

Sources: TechCrunch (Sep 2026; 150M+ as reported, company confirmed licence and passport data), company disclosures and NHK (Japan), TechCrunch / NBC (Oct 2025), Reuters via Fortune (Jul 2025), Latitude Financial (Mar 2023).

THE PROBLEM · IT IS ABOUT TO GROW

Laws now demand age and ID checks — and most checks still mean uploading your ID.

UK · JUL 2025

Online Safety Act: "highly effective" age checks for adult sites

AUSTRALIA · DEC 2025

Minimum age 16 for social media accounts

US · 2026

About half of the states mandate some form of age gating

JAPAN · APR 2027

Remote KYC moves from photo uploads to IC-chip reading

More checks × stored copies = more breaches.

$4.44M

average cost of one data breach, worldwide (2025)

Sources: Ofcom; eSafety Commissioner; Jumio age-verification law snapshot (2026); JAFIC (NPA); IBM Cost of a Data Breach Report 2025.

CLOSE-UP · JAPAN, SEPTEMBER 2026

A car-sharing service lost its members' ID cards.

1.6M

images of driver's licences and other ID documents uploaded at sign-up

6.6M

member accounts: name, address, date of birth, licence data

30.6M

people hit by leaks at Japanese listed companies in 2025 alone

Sources: company disclosures (Sep 2026), NHK (Oct 2026), Tokyo Shoko Research (Jan 2026). Police are reissuing affected licences.

Data you never collect cannot leak.

TODAY

`collect → store → leak`

Every service keeps a copy of your ID.

WITH P!PON

`verify → forget`

The service checks a proof and keeps nothing.

This is the turning point: services will stop holding personal data wherever they can.

WHY START IN JAPAN

Japan is moving identity checks from photos to the chip.

Apr 2027

Remote KYC under the anti–money-laundering law drops photo uploads in favour of reading the IC chip.

104.5M

My Number Cards held (84.5% of the population). Each chip can carry a certificate signed by the national authority, J-LIS.

Sources: National Police Agency (JAFIC) Q&A on the amended ordinance; Ministry of Internal Affairs and Communications, cards held at end of Sep 2026.

IC chips solve authenticity.
Zero-knowledge proofs solve privacy.

My Number Card

Certificate signed by J-LIS. Signs a fresh challenge.



Phone

Checks the signatures inside a zero-knowledge proof.



Website or Solana

Verifies the proof. Gets a yes.

The service learns

Over 20 · lives in a prefecture · hasn't claimed before

It never sees

Name · date of birth · address · ID image · My Number

WHO IT IS FOR

Every service that asks for an ID only to learn one fact.

Online stores
Alcohol, tobacco, adult content

Sign-up checks
"A real, unique person"

One per person
Drops, airdrops, giveaways

DAO votes
Sybil-resistant governance

AI agents
Acting for a verified human

Not legal KYC
It's for everything else

WHY SOLANA

A shared "one per person" registry that no company has to own.

~144k compute units: Groth16 verify (alt_bn128), nullifier and NFT mint in **one transaction**

One per person, no list of people

Each card leaves a per-gate nullifier in a PDA. A second claim fails on-chain.

A trust list anyone can read

Gates hold the trusted J-LIS CA keys on-chain. Our Web2 store reads the same list: no API key, no vendor.

Next: attestations and agents

Solana Attestation Service for reusable credentials; x402 so an AI agent can show a verified human stands behind it.

Live on devnet: zk_gate program with real-card and demo gates · see the technical demo

ROADMAP

From one card in Japan to every chip, for people and their agents.

NOW · OCT 2026

Working sample

My Number Card: age + prefecture + one per person

zk_gate on Solana devnet; browser verifier for Web2

iPhone on-device proof (9 s); Mac + USB reader

Tested end to end on a real card

NEXT · BY Q1 2027

Product for stores

iPhone app in TestFlight, then Android

Drop-in checkout SDK for online stores

More facts: age lines, city, recent certificate

Solana Attestation Service; multi-party setup, audit, mainnet

AI AGENTS · 2027

A human behind the agent

Delegate a scoped, revocable "verified human, 20+" credential to your agent

Proof attached to x402 payments for age-restricted goods

Per-human limits via nullifiers: one person, N agents

Tools for agent wallets and MCP

GLOBAL · 2027+

Every ID chip

ePassports to the ICAO 9303 standard: chip data signed by each country's CA

Other national eID cards with signed chips

Same proof, same Solana registry, any country

Revocation through certified operators

ROADMAP · PRODUCT DESIGN · WEB3

Scan once. After that, just sign with your wallet.

ONCE · ABOUT A MINUTE

1. Tap the card; the proof "20+, one per person" is bound to your wallet.

2. zk_gate verifies it on-chain and issues a **non-transferable attestation** to the wallet (Solana Attestation Service), with an expiry.

EVERY TIME AFTER · SECONDS

3. A service asks the wallet to sign — no card, no password.

4. It reads the attestation on-chain: issued by zk_gate, not expired, held by this wallet.

Bound to the wallet

Non-transferable: it can't be sold or lent.

It expires

Re-scan yearly or when the certificate renews; address facts sooner.

One per person

The nullifier stops a second wallet for the same card.

Your choice of linkage

One wallet everywhere is convenient; separate wallets keep services apart.

Planned. Today's demo proves on every request; the attestation turns that one proof into a reusable credential.

ROADMAP · PRODUCT DESIGN · WEB2

Verify at sign-up. After that, just log in.

ONCE · AT SIGN-UP

1. Tap the card; the proof is bound to the store's sign-up session.

2. The store verifies it and saves on the account only: **"20+ verified on (date)"** and its store-only pseudonymous ID.

EVERY TIME AFTER · USUAL LOGIN

3. Log in with a password or passkey, as today.

4. Checkout reads the flag on the account: alcohol can be sold, no ID asked.

Nothing to leak

A breach exposes a yes/no flag, not a licence image.

One account per person

The pseudonymous ID blocks duplicate sign-ups.

Re-check when it matters

Yearly, or before a high-risk order, by the store's policy.

Or reuse across stores

A store can also accept the Web3 attestation: no scan at all.

Works with today's verifier: the store keeps the result of one proof instead of a copy of the ID.

STATUS

Working today on devnet, end to end with a real My Number Card.

Next: stores, AI agents, then every ID chip — see the roadmap.

Stop collecting IDs.
Prove the fact.



TECHNICAL DEMO · ARCHITECTURE

No issuer. The national CA is the root of trust; the phone does the proving.

Layer	Built with
Circuit	Circom · Groth16 · 1.49M constraints
Prover	mopro + rapidsnark on iPhone · Rust CLI on Mac
Card I/O	Core NFC (iPhone) · CryptoTokenKit / PC/SC (Mac)
On-chain	Anchor program zk_gate · alt_bn128 · Metaplex Core
Web verifier	snarkjs in the browser · CA trust list read from Solana

INSIDE THE PROOF

Four checks in private. This sample reveals two facts.

1. J-LIS CA signature over the certificate (RSA-2048, SHA-256)

2. The card's signature over a challenge bound to the wallet or browser session

3. Date of birth + N years ≤ today

4. Address starts with the chosen prefecture

Revealed in this sample

Age ≥ 20 · Prefecture

Plus the plumbing: CA key hash, a per-service nullifier, challenge, date, scope.

Room to grow — the certificate also holds name, date of birth, sex and full address, so the same pattern can prove (not built yet):

- Other age lines: 18+, 65+, under 25
- City or ward
- Sex, only where a service must know
- Name matches the account (hash, not the name)
- Certificate issued within N days

FLOW 1 · WEB2 · ONLINE STORE SELLING ALCOHOL

How the store learns "20+" without seeing your ID

STORE

1 · Checkout asks

Shopper taps "Verify age with p!pon". The store makes a one-time session ID and shows it as a QR code (a link on mobile).

PHONE

2 · Request opens

The p!pon app shows who asks and what: "Kura Online Store · Age 20+".

CARD

3 · Tap the card

After the signing password, the chip signs SHA-256(session ID ‖ nonce) and gives its J-LIS-signed certificate.

PHONE

4 · Prove on device

Groth16 proof (~9 s): CA signature, card signature, age ≥ 20, prefecture. The certificate never leaves the phone.

STORE'S BROWSER

5 · Verify, ~200 ms

Proof valid · same session (no replay) · CA in the trust list on Solana · dated today · 20+ · scoped to this store.

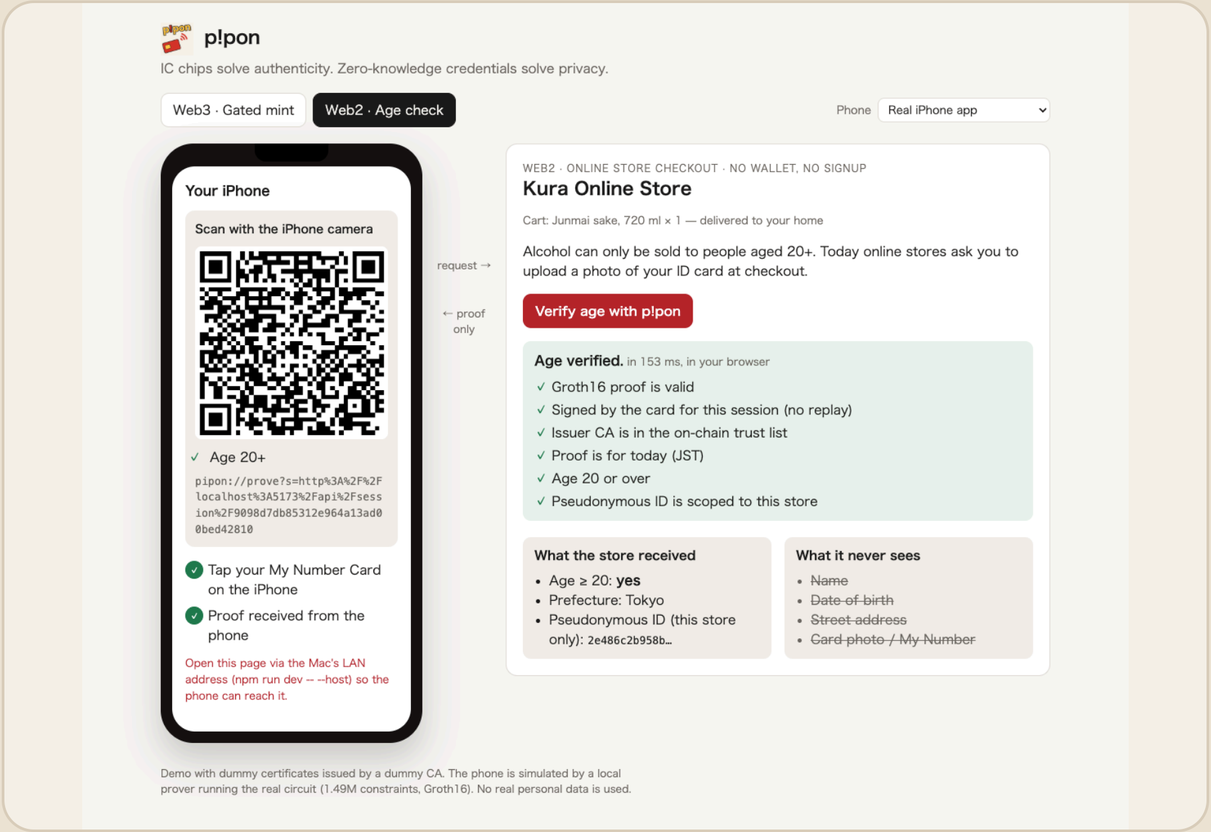
Store keeps: age 20+ = yes, prefecture, a store-only pseudonymous ID.

Never receives: name, date of birth, address, ID image, My Number.

Once per account or order, from home — not a card scan at a shop counter.

DEMO 1 · WEB2 · ONLINE STORE CHECKOUT

No wallet, no signup: the store gets a yes.



An online store selling sake: age 20+ needed at checkout

Verified in the browser in ~200 ms; issuer CA checked against the trust list on Solana

Store learns: 20+, prefecture, a pseudonymous ID for this store only

FLOW 2 · WEB3 · ONE-PER-PERSON MINT

How Solana enforces "20+, one per person"

WALLET

1 · Open the gate

The gate account on Solana holds the rules: minimum age, prefectures, trusted CA keys, scope.

PHONE

2 · Bound to the wallet

The challenge is SHA-256(wallet address ‖ nonce), so the proof only works for this wallet.

CARD + PHONE

3 · Tap and prove

The card signs the challenge; the phone makes the same Groth16 proof as in Web2, with the gate's scope.

SOLANA · zk_gate

4 · Verify on-chain

Groth16 via alt_bn128 syscalls; CA, age, prefecture, date and the signer's wallet checked against the gate.

SOLANA · zk_gate

5 · Mint or reject

New nullifier → PDA created, Core NFT minted, same transaction (~144k CU). Nullifier exists → "Already claimed".

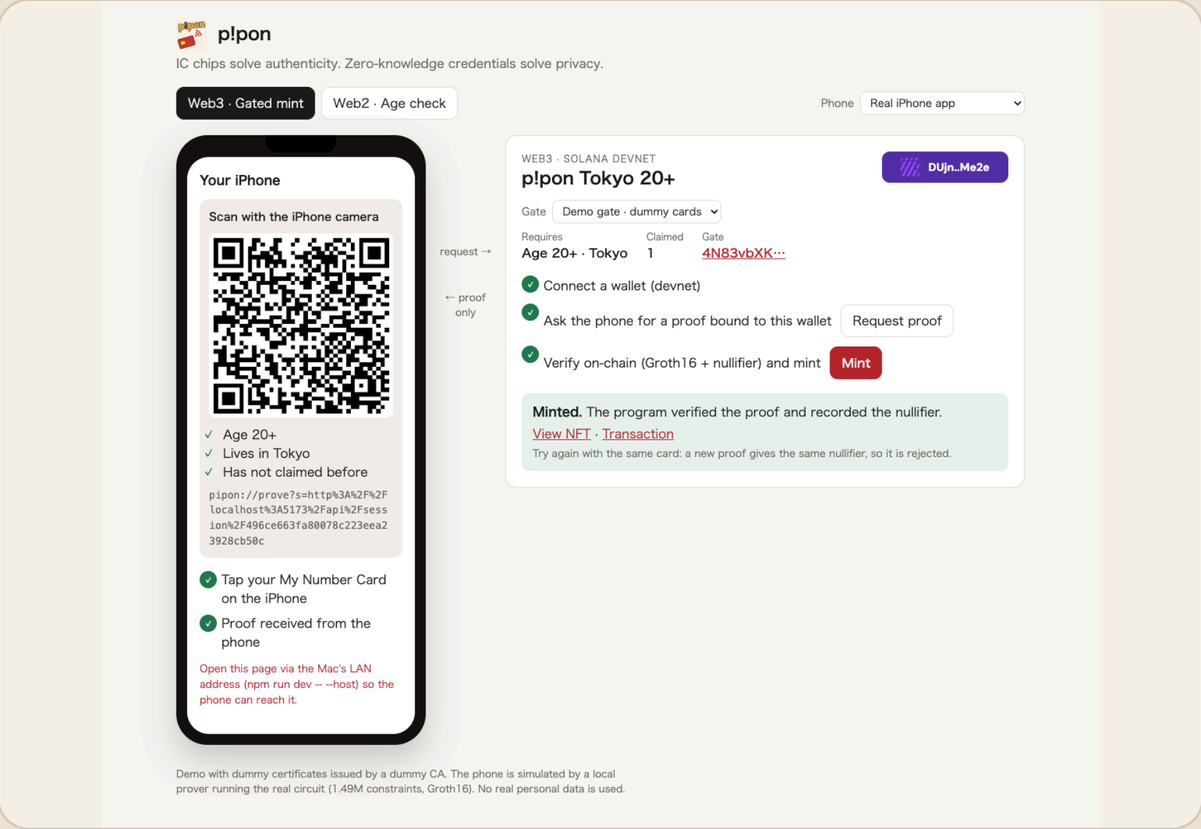
Public on-chain: the proof and its values — CA key hash, nullifier, date, age threshold, prefecture code.

Never on-chain: name, date of birth, address, ID image, My Number.

The nullifier is per card and per gate: another gate gets a different one, so wallets can't be linked across services.

DEMO 2 · WEB3 GATED MINT

One transaction: verify, record, mint.



Proof bound to the wallet

Same card again → **Already claimed**

Card from another prefecture → **rejected by the gate**

TESTED ON A REAL MY NUMBER CARD

Same flow, real J-LIS certificate.

3.5 s

proof on a Mac mini, card in a USB reader

9 s

on-device proof on an iPhone 15 Pro Max

2 / 2

claim verified on devnet · second claim from a new wallet rejected

Devnet test gate (scope 3), 2026-10-07 · tx c623tn…zvYv7 · no personal data shown or stored

LIMITS AND NEXT

Honest about what's left.

Limits today

No revocation check yet: revocation lists are only for certified verifiers.

Single-contributor trusted setup: demo only.

How the roadmap closes them

Revocation through a certified platform operator.

Multi-party trusted setup and an audit before mainnet.

iPhone NFC in TestFlight (paid signing team pending).